Day 4: Hands-on Subnetting

Q1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

- Subnets needed = 4. 2^2 = 4, so 2 bits are borrowed from the host part.
- New prefix = 24 + 2 = /26. Subnet mask = 255.255.255.192.
- Block size = 256 - 192 = 64 addresses per subnet, with 62 usable hosts.

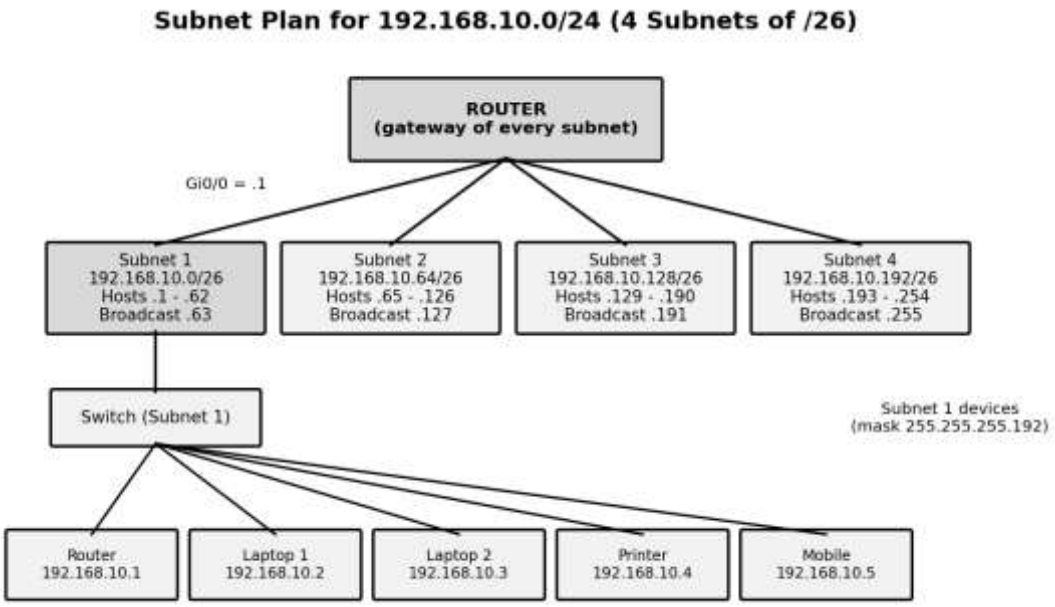
Subnet	Subnet mask	Network address	Broadcast address	Usable host range
Subnet 1	255.255.255.192	192.168.10.0	192.168.10.63	192.168.10.1 to .62
Subnet 2	255.255.255.192	192.168.10.64	192.168.10.127	192.168.10.65 to .126
Subnet 3	255.255.255.192	192.168.10.128	192.168.10.191	192.168.10.129 to .190
Subnet 4	255.255.255.192	192.168.10.192	192.168.10.255	192.168.10.193 to .254

Q2.
For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Make sure not to use the network or broadcast address for any device.

Device	IP address	Subnet mask	Default gateway
Router (gateway)	192.168.10.1	255.255.255.192	-
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile	192.168.10.5	255.255.255.192	192.168.10.1

All 5 addresses are between 192.168.10.1 and 192.168.10.62, so they are valid host addresses. Addresses .6 to .62 are still free for new devices.

Q3.
For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Make sure not to use the network or broadcast address for any device.



The router is the gateway of all four subnets. Subnet 1 is shown in detail: the router, two laptops, the printer and the mobile are connected through a switch.

Q4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Think of the network 192.168.10.0/24 as one big building with 256 rooms (addresses). Subnetting divides the building into smaller equal floors, so that each group of devices has its own floor.

Step 1: Understand what /24 means. /24 means that the first 24 bits (the first 3 numbers) identify the network and the last 8 bits identify the devices. So there are 2^8 = 256 addresses in total.

Step 2: Decide how many subnets are needed. We need 4 subnets. We find the power of 2 that gives at least 4: 2^1 = 2 (too few), 2^2 = 4 (enough). So we take 2 bits from the host part.

Step 3: Find the new prefix. The old prefix is /24 and 2 bits are added, so the new prefix is /24 + 2 = /26.

Step 4: Find the subnet mask. /26 means 26 ones: 11111111.11111111.11111111.11000000. The last octet 11000000 = 128 + 64 = 192. So the mask is 255.255.255.192.

Step 5: Find the block size. Block size = 256 - 192 = 64. This is the number of addresses in each subnet, and also the jump between one subnet and the next.

Step 6: Find the hosts per subnet. The host bits are 32 - 26 = 6, so 2^6 = 64 addresses. Two addresses are reserved (the first is the network address and the last is the broadcast address), so usable hosts = 64 - 2 = 62.

Step 7: Write the subnets. Start at 0 and add 64 each time: 192.168.10.0, 192.168.10.64, 192.168.10.128, 192.168.10.192. The broadcast address of each subnet is one less than the next network address: .63, .127, .191, .255. The usable hosts lie between the network and broadcast addresses.

Step 8: Assign IP addresses. Choose Subnet 1 (192.168.10.0/26). Give the router the first usable address 192.168.10.1 (it is the gateway) and give the other devices the next addresses .2, .3, .4 and .5. Every device gets the mask 255.255.255.192 and the gateway 192.168.10.1.

Step 9: Check the plan. Make sure no address is repeated, no device uses a network or broadcast address, and all addresses of one subnet are inside its range. Then ping between the devices to check that they can reach each other.

Quick reference: subnet masks

Prefix	Mask	Block size	Total addresses	Usable hosts
/25	255.255.255.128	128	128	126
/26	255.255.255.192	64	64	62
/27	255.255.255.224	32	32	30
/28	255.255.255.240	16	16	14